

ActInf Textbook Group ~ Cohort 3 ~ Meeting 19

(Chapter 8, part 2)

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Duration: 56:27 | Uploaded: Unknown Transcript method: auto_caption

foreign it's July 18 2023 and we're in our second discussion on chapter eight so where would anyone like to begin or any general remark on chapter eight what I can say something I added a couple of questions to the list at the very end on box 8.1 but we can go through some other stuff first if you want sounds good um all right let's look at the first question yes perhaps if I put some background in of my understanding so so my understanding is we have uh well I think if we go to there's Rafael Boyach or Bogach's paper a tutorial on free energy uh which starts off with looking at a single neuron with an input uh with an input distribution of a particular mean and and variance yeah and and he then builds that up uh to the point where you've got a multi-layer system with multiple nodes and the uh the the synapses represent uh means and variances I think there's we're not seriously suggesting that that synapses actually represent particular means and variances or whatever but I think there's the idea that collectively a population of neurons is going to encode uh mean values and variances um and position is is inverse variance of course uh I suppose my other frame of reference is gaussian mixture models where you've you know you've got gaussian's characterized by uh weights or amplitude mean and and variance something going on to box 8.1 I'm confused that there seems to be a way of treating a Precision of a as a magnitude as an amplitude um or changing that changing the gaussian uh perhaps of buying reducing the variance increase in the Precision is is going to increase the peak to make it more prominent uh which seems a strange way of doing things yeah exactly so so for me well precision and attention are not synonyms and attention seems to be a or something at the neuromodulation time scale as opposed to synapting weights at a much slower time scale or General influence with with usual synaptic or some neurotransmitters uh so yeah it it doesn't make sense how uh well a lot of a lot of the uh of that box your mute um yeah Ollie or anyone do you want to give it that or okay so here the Precision being described is the Precision to the sensory input and so equivalent to attention to in the sense that um if the uh and reflected by the a matrix so if uh

the incoming sensory data are inferred to be accurate then they will update the hidden state more if the sensory data incoming were not believed to be accurate low Precision low attention then their then observations wouldn't update hidden States I don't think that that necessarily refers to the psychological concept of attention but is your question [Music] about well yeah in terms of significance you could have something that was that was more important or more worthy of attention at a particular time even though it had less but a belief was held with less precision what would you mean more important like more relevant for survival importance more relevant yeah I think I so I I say it as from uh a gaussian mixture model point of view I suppose so you could have but let's let's take I think I'll put in the questions of if if you have if you have two gaussians let's say you've got a a single sense input uh of measuring the wavelength of light and you're presented with apples and pears and so pears cover a fairly narrow green spread of the green so you've got a very a very sharp low variance distribution there and your apples spread right across red to Green but that your input your environment might actually have uh a lot more apples than than pears so even if you're in the in a uh if you sense something that was within the the pears area of the spectrum it still might be more likely that that you've had an apple that's exactly the prior like and that that's um factorized away from the attention to the sensory input the D is coming into the s from the left side and the attention to sensory input is coming up from the bottom so you're able to attend to sensory input or not but but it is very it is very I believe useful to consider what happens here in in terms of um the variational estimation where there's no action we're not choosing which thing to look at yet but still in this mapping the the fruit identification region of the brain the rate code would represent the posterior on Apple to pear zero if it's an apple one if it's a pear and then the variance would represent the uh the Precision on the sensory input mapping to it such that a high precision High Precision belief which is to say low Precision on sensory input would have high inertia as a belief I think that's accurate I think it part of the part of the issue is that um Precision can be on any variable like uncertainty could be on any variable so it has to be clarified Precision on what and this is on sense Precision on and I think a question that Sanjeev answered in and perhaps the most recent uh meeting attention waiting rescales the um attention to different variables is what dictates the influence that they have on the overall free energy gradient descent if you have two variables and they both are equally attended to then the free energy is a 50 50 compromise such that there's going to be a flat manifold where one of them is getting better by one point one's getting worse by one point whereas if you're attending to one of them but 10 times more than the other so it's 10 times more Precision on

that belief each increment of free energy minimization you'd be willing to accept 10 decrements on the other one it's the attention blends the the role of each parameter being optimized on the overall scalar gradient descent yeah I yeah I've still got a bit of a block on the on that I'll go away and if I think about that but I think what you said just leads naturally on to my other point so I so as I see it as as you say the uh that's that rebalancing of the attention so if if you if we have a tension given to uh extra set of input then you know you're sensing the outside world you're treating that as important if you then shift to proprioceptive attention that's that's more likely to to lead or that would lead to action or that there's there's the whole thing about sensory attenuation why why does sensory attenuation happen is it a random thing well there's there's the splitting across and there's the there's a a neuroscientific model framework and then there's also actual we're trying to do some engineering stuff here but on the the neuroscience well yeah generally are we saying that there is some oscillation uh flip-flopping between uh sense and effectively action or is it just something that's happening randomly that's that things will drift and get to it an action phase or sensory balance yeah I think there's a lot of pieces there um so systems don't have to have sensory attenuation if having a constant uh uh regime of attention it is simply how the system is defined uh by Fiat or at some level of generative modeling for some variable there may not be any reason to to propose this cognitive phenomena um however whether you look at isicating and visual input where we have like Precision High Precision on the sensory input when we're fixed and then we pull off the gas when we move so that we suppress input while the visual field is very shifted and then re re-engage so whether we think about like the icicade attention to visual input or if we think about um the uh motor control example in box 8.1 which is exactly analogous it's just interceptive instead of x terreceptive where there's a suppression of sensory um support action selection instead of um belief updating in both those cases a kind of adaptive emergent cybernetic pattern is this oscillation of attention to enable the benefits of um High attention to sensory input but then also be able to transiently engage another mode where input to sensory uh uh data is lowered so I think that just turns out to be a solution or an approach to navigate the the the trade-offs because you wouldn't want to just have high attention or only low attention so then then the navigation solution oscillates between them so this is so we're saying that this is just something that happens to be found in real grains or as fit into a model of of real brains it's not something that is inherent within active inference it's not something that appears in the in the discrete time uh domain is it um I I guess you could model sensory attenuation in discrete time but correct sensory attenuation is not unnecessary

part of active inference agents so why is this useful because this pattern helps us understand the cognitive similarities between speaking and listening where the bird song example is a famous case of sensory attenuation like you expect the song to be playing so as long as it's playing you listen and then when it goes away then now you take action and then while acting your Precision decreases until you eventually see it acting again so turn taking conversation is a cascade of motor control with proprioceptive the cognitive patterns can be identified and those are accounts of complex phenomena that are first principles and don't appeal to reward or reinforcement they appeal to some specific computational phenomena that Maps between the neurobiology Loosely like you brought up with a mean in the variance and the statistical parameterization and then that model generates unique explanations and predictions and say oh well maybe this um dyskinesia is a challenge of initiating movement and this one's a challenge of terminating movements yeah yes and there's there's a time with various mental illness as well isn't there so schizophrenia title with sensory generation it kind of reminds me of the uh the work by Adam Saffron of Albus because Friston and Carhartt Harris had this Rebus paper relaxed beliefs under psychedelics and then there was like this other proposal was like sevis strengthened beliefs under psychedelics and then Adam was like well it's kind of just altered oh that's really General but it's just increasing the Precision on this one right then decrease the Precision on something else so there's not just a Precision variable it has to be a little bit more unpacked with Precision on what beliefs because we could talk about beliefs on um since uh States about beliefs on latent States believe some action it's all belief updating it's all Bayesian beliefs and can you think is there any reference that I perhaps I should read that um might get me thinking about Precision in a better way maybe the bird song um Kristen and Frith bird song example I can also maybe with Sanji if we can ask and see his simulation seem very amenable to it yeah I think possibly because I've got it fixed in my mind the idea of synapses encoding variants uh whereas as you say variance is there's this variance uh much more broadly were in the concept of making inferences uh posteriors that's where I don't really have a good handle I think this is a you know there's a barrier I have a barrier so how do I get through the barrier yeah that's kind of a hard like a topic and it just reminds me of the end of seven like we call D the prior but of course once you escape the most trivial pedagogical generative model then it makes more sense to be clear this is a prior on hidden States because you're also talking about a prior or hyper prior on other variables and also that was like kind of the motivation for the generalized notation notation so that we could which was originally called generalized subscript notation because we wanted to be able to talk about like a

preference on preference or a policy on preference or a prior on preference because when you're in the nested models you have to clarify which term you're talking about okay yeah so I'm I'm pretty much happy with the rest of the chapter it's it's that it's just that one box that I need to get around so yeah so like you said the Birdsong thing so I've read that in the past that that makes sense but it's this ambiguity I have around the word Precision um and how it well that that one phrase of precision and attention being synonyms is that that I've got the block on me let me try one one more angle on this so we're sitting up and sitting down example from the sensory attenuation so let's just say that our proprioceptors give us like 90 confident information when we're in that state but when we're moving it gives us um 50 50 so it's totally ambiguous but then it's 19 90 10 once we're in the seated or the standing um state we have a true hidden state which is our actual posture and then um let's just say that we're getting seated input because we're seated but because of a cognitive decision now we prefer standing input now if the pragmatic value were scaled super high then you would take the policy to stand up because the pragmatic value would just overwhelm everything else however if the pragmatic value is not scaled super high then there's a little bit of like a frustration it's your preferred state but not so preferred that it's overwhelming the the belief updating to be anchored closer to the um sitting down so you have kind of these two peaks you could be getting you could be you know drawing from this peak of observations sitting down or you could be drawing from the other Peak um and then um even if this even if the other Peak is slightly higher the optimization still holds to the left side and then like by increasing the variance reduce your precision the in the in that ambiguity from the then the preferred State then you can select action to stand up and kind of it's like temporarily flattening the landscape so that you can make a move without needing to um want it more yeah I think something you said at the beginning that was very significant I think because I'm I'm thinking when we're talking about Precision it's Precision of beliefs of whereas there's also Precision on uh in states and you know between priors and new input uh it's that balance see what else happens okay

lockableterra Thomas part um explains some of this like the the historiosity of this lockableterra was used to uh describe structured sequential activities in the continuous time setting okay learning a functional s that plays the role of a free energy this could definitely be unpacked a little but it seems like this is a continuous time free energy based method for gradients based optimization of mean and variance of the LaPlace approximation well it seems like I could use a little bit more context I'm I'm right in thinking so so if if we visualize a normal distribution and we're saying we increase the Precision uh you know we can think of

that as the shape the change of shape of the distribution uh so that the peak goes up so it's it is that that's the right way of thinking about it we're changing the shape of the normal distribution so hence increasing increasing the Precision so reducing the variance is the same thing as increasing the the peak at the at the mean value which is the mode as well yeah yeah yes it sharpens it when it's more attended to and this is sort of like the two um the two faces of the KL Divergence is mean seeking versus mode seeking and attention modulation influences which one of those it goes to as well as where that optimizations um starts in the landscape because if it's too attentive if it's too attentive to get out of a local Optima then it becomes functionally mode seeking because whichever mode it gets to it'll just think okay I'm doing as well as I can because also LaPlace doesn't um the gaussian has support everywhere but a LaPlace is a downward facing parabola so it doesn't have infinite support whereas so it's not exactly a gaussian mixture model and then the mode seeking it just tries to find whatever the highest point is and then fit a gaussian around it versus the mean seeking tries to cover as much of the distribution and match the center of gravity with the central tendency so in the case of the sitting up and standing standing um you you're you're usually wanting to be pretty precise about your body's position so you're mode seeking you're mode seeking you're locally you're sitting down then it's like the attention relaxes that moves the kl to be more like mean now it's like 50 50. 50 your prior half the time you're standing half you're seated and the sensory data are ambiguous and you prefer to be standing so then that preference breaks the symmetry and then you can re-sharpen your attention back when you're standing so then the KL can have kind of this like end to end um reduction of diversions and the only knob that you had to change was the attention now how to adaptively tune attention that's a question but that's why it's it's um interesting conversion to Evolution that we see the same like cyclic attention strategy in the cicade and in the um standing example so we see that as a repeated heuristic umdp chapter 7 has largely superseded the use of generalized lockableterra and active inference that's kind of interesting because that's like saying the discrete time has kind of taken over from The Continuous time I've made the point before so a lot of those sort of classic examples were including the bird song in the SPM 12. is is based on the SPM ADM active Dynamic expectation maximization model so that's that's all within the continuous well this continuous time frame concept as opposed to pomdp so it's it's like that's yeah it has been superseded because all of the com so much of the conversation now is about bomdb eventually we'll we'll have every model annotated in literature so we can look at like the trends and the um which ones were used but suffice to say lockableterra I don't see this coming up recently

and continuous times are definitely not as frequent as POS mdps I mean Pi mdp only describes continuous and it's just so much closer to the setting of uh discrete time explicit planning counterfactuals uh reinforcement learning reward like the discrete time formalism is a lot closer to it okay I suppose yes more more useful for sort of high level sort of cognitive functioning as opposed to the basic movement in an environment but that's kind of the cool like Internet of Things question like what if you could have a you know a wing um or like a water you know something and then all that it was really like endowed with was heuristics around attention modulation maybe that only takes like two lines or something and then like with this heuristic ground then it can do like a in a continuous sensory capacity whereas you could TR you could train a and do all these big like Matrix calculations I mean that that's sort of the question where do these um different models have difference computational advantages or explainability advantages yeah okay thanks yeah oh yeah these are great questions also the uh the recent model stream nine it'll I'm just reloading it with Aspen Paul um here it is he presents a computational the only other time we've seen computational complexity estimates was with uh branching time active inference I thought this was super fascinating so here's um so classical octave s-i sophisticated inference time step 30 horizon or two Horizon DP dynamic programming um aif one is um is just single timestep active so 30 times step sophistication 10 to the 68 complexity cognitive complexity and just like how I just found it really interesting how the sophisticated inference blow up is so high and and uh he's using dynamic programming based methods to learn smooth priors on preference so I think there's a lot of difference there's a lot of different little motifs that might have very effective performance in a certain setting like in the setting of a camera that moves isocating and blinking bite converge to being an Adaptive strategy so finding out which generative models and phenomena parsimoniously describe these different natural systems might be very helpful for them being able to compose them together that understand okay where do we need thank you Ali talk to you later where do we need different difference models I think I asked him like can you do like the full can you do like this approximation in this part of the nested model in that one in this part and he said yes so you could even Implement two different approximations for different parts the same generative model I think okay then generalize synchrony okay here's where the bird song comes back into play bird hears the song there's no need to generate it if the bird predicts a song that is not heard it must start singing learning here it's uh described more in the paper I assume that it is learning according to this box this leads to turn taking sometimes phrase to seeing from the same hymn sheet yes uh it is

therefore it's not really like a model of improvisational communication or necessarily even communication as information transmission they're literally just it's like there's something that has to be carried and they just like one of them carries it until the other one picks up so it's like it's a very basal aspect of communication because the model is really just trying to describe to turn taking to generate bird song requires a reduction in the Precision of predictions about the consequence of action I'm not sure why that is why couldn't you take action and also increase your Precision about the consequences of your actions that'd kind of be like action um I mean might even be connected to something like OCD where action that that action that when you take it blurs your precision is kind of like self-terminated because eventually you'll just wobble out of that behavioral attractor whereas if there's a behavior that is that is increasing your so you're attending more and more to the sensory outcomes that that action is seeking then I feel like that action becomes like Tighter and Tighter um like in a tractor but something like like um I mean even though a lot of times different uh neuropathologies and stuff come up I think the reality is a lot of these are really sophisticated multi-level cognitive phenomena and so it's just kind of pointing in that direction but it's not really it's not it's not that's the same thing generalized synchrony again seeing from the same hymn sheet they have learned to share the same narrative I think if you consider the song to be a narrative and also open question is like how much is it is already like baked in and then it's being fine-tuned versus it's not like they're starting with two different singing Concepts and then actually learning the whole thing but just okay then Hybrid models single models that contain both continuous and discrete variables in this case the lower level is continuous time $x \times \text{Prime} \times \text{sub Prime}$ the higher level is um discrete time past present and future this is the same family of model that Ryan Smith at all explored in live stream 46 active inference does not contradict folk psychology where the lower level is the motor active inference and then the higher levels the decision making active inference there's supposed to be an edge here uh message passing uh procedures exist for graphs that have discrete and so they're kind of brought up separately just because of uh pedagogy but ultimately it seems like there's no issue with mixing the the mapping between discrete and continuous happens by basically giving discrete State spaces uh integer values in some continuous space State space you could associate the coordinates with the discrete location but then also have it be a continuous space so then you I guess can have the discrete time model and the continuous model in the same um being like kind of described in one from the discrete side as zero one and then in the continuous one from continuous space from zero to one okay this is a little bit of the um this is an icated model probably this is a

useful paper to look more into 16 neural populations four locations and four time steps one two three and four but I don't know why they stay active I don't know why they don't drop back down well and here's what what's happening in this is the person is being told to look at the cross a DOT is being shown at the top of the screen there's some 200 millisecond delay that represents like visual processing then they really they accelerate here's their point of maximum speed the accelerate and then decelerate and fixate on the queue where the cube was and each of these four traces represents beliefs but again I'm not sure why why it's not just like one comes up and then it goes down the second comes up it goes down okay mixture models gaussian mixture models like you brought up earlier what do you think the relevances of the gaussian mixture here Neil because there's reference to sort of Delta functions here so precise well totally precise prize there so I'm not sure where the gaussian comes in other than that bottom equation the normal distribution the priors on the mean and the variance are precise so we're basically just looking for a point estimate on mean and a point estimate on variance not looking to have a secondary uncertainty about mean and in a secondary uncertainty about variance that's the Delta part and then I guess the I subscript is the number of mixtures and maybe different observations are being allocated or learned the difference mixtures categories so the optimal you know if you had two if you had one mixture it would be the same as a gaussian if you have three mixtures then you'd be trying to parameterize three different gaussians with different means and variances and then assigning each data point to one of those mixtures and then the question of which which number of mixtures of groups should you have is approached through Bayesian model reduction Bayesian information Criterion eight one Birth song ocular motor condition reflex smooth eye Pursuit psychosis illusion cicades auction observation attention hybrid model self-organization yeah interesting and kind of very sparse and brief chapter I feel like it is very short and just barely touches on a few pieces but I guess provides information where there's other continuous time models what do you think yeah nice it was very good chat so what this this is this is more where my interest lies as soon as you might have guessed I want to continue a Time side of things um I just have that one difficulty around the Box having accepted what's in the box everything else follows fine it's it's yeah it all hangs together quite neatly but yeah there's various things have been said I need to go away and digest I'm just just like you're saying a point estimate on variance um whereas I'm thinking of perhaps and confusing and getting estimates and variants confused you know the biggest variants included in the estimate um right and we we variance is like the mean on our variance estimator and and uh usually we don't go two levels deep like usually the the

standard uh deviation there's not an SD on the SD usually that's just the point estimate on mean and variance but then it's like but the variance of the mean isn't that just the variance how is the variance different different than the Precision on the mean yeah plus I I mean let's think about the class you know the people are 10 feet tall plus or minus five so variance is five by virtue of the actual variation in height but the Precision on the mean if we sampled a million creatures might be very low but we might have a very confident estimate it's 10 plus or minus 5. or we might have a very imprecise estimate that it's 10 plus or minus 1. and then in the variational gradient descent scheme attention relates all the variables so that the gradient descents strikes a balance but that just begets the question how does the attention get trained so that in a multivariate optimization variables are being traded off appropriately including situations where there's a dynamic reallocation of of optimization Focus hmm you know how how is how is the whole system well how are those parameters of the the um the chaotic attractor tuned so that it does um it does follow contain trajectories stable rather than just going off and disappearing down getting stuck into into in one attractor rather than another yeah well for uh for a system to be able to be uh to descend into an attractor and to escape it kind of has to be pushed on the edge of chaos perhaps some dynamic or oscillatory component otherwise it just gets attracted towards attracted to but okay cool thank you well tomorrow's the first math learning group or or so so people are welcome to come to Discord if um if they want but otherwise see you all next time that's great thanks Michelle bye